Zihao Zou

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EDUCATION

University of North Carolina, Chapel Hill — Ph.D. student, Department of Computer Science — 2024 – Present · Chapel Hill, NC

Washington University in St. Louis — M.S. in Computer Science, McKelvey School of Engineering — 2021 – 2022 · St. Louis, MO

University of California, Irvine — B.S. in Computer Engineering, College of Engineering — 2016 – 2020 · Irvine, CA

RESEARCH INTERESTS

My research focuses on developing robust and efficient frameworks for computational photography and scientific imaging with GenAI. In particular, I work on high-speed, high-quality 3D imaging through multi-modal sensor fusion and optics design. I also have broad interests in imaging inverse problems, generative AI, and metalens design.

Keywords: Computer Vision, Computational Imaging, Large-scale Optimization, Machine Learning, Event Camera, TOF Camera, Camera Fusion, Diffusion Models for Imaging, Metalens Design.

PUBLICATIONS

- **Asynchronous Multimodal View Synthesis with Probabilistic Temporal Encoding.** *SIGGRAPH Asia 2026.*
  Z. Zou, J. Liu, H. Lian, N. Thadishetty, P. Chakravarthula.
  Developed a diffusion transformer with probabilistic temporal encoding to fuse asynchronous RGB and event streams for temporally consistent novel-view synthesis.
  Project Paper

- **Why Slow Feels Fast and Fast Feels Slow: Evaluating and Predicting Speed Misperception.** *ISMAR 2025.*
  X. Peng, C. Groth, D. Jimenez Navarro, **Z. Zou**, Y. Zhu, A. Serrano, K. Myszkowski, Q. Sun, P. Chakravarthula.
  Investigated how visual scene properties bias perceived speed and developed a predictive model validated in immersive VR environments.

- **Implicit and Explicit Volumetric Representations for Sensor Fusion.** *Asilomar 2025.*
  Z. Zou, Z. Qu, S. Jayasuriya, A. Pediredla, P. Chakravarthula.
  Explored implicit and explicit volumetric representations for integrating complementary sensor measurements.

- **High-Speed Dynamic 3D Imaging with Sensor Fusion Splatting.** *Transactions on Graphics · Under Review.*
  Z. Zou, Z. Qu, V. Boominathan, A. Pediredla, P. Chakravarthula.
  Fuse RGB, depth, and event cameras together under a unified framework, enabling high-speed and small-baseline 3D/4D imaging and inverse rendering.
  Preprint

- **Event Fields: Capturing Light Fields at High Speed, Resolution, and Dynamic Range.** *CVPR 2025 · Highlight.*
  Z. Qu, **Z. Zou**, V. Boominathan, P. Chakravarthula, A. Pediredla.
  Combined RGB camera and event camera with special optical designs, enabling high-speed, high-resolution, and high-dynamic-range light-field imaging.
  Paper

- **FLAIR: A Conditional Diffusion Framework with Applications to Face Video Restoration.** *WACV 2025.*
  Z. Zou, J. Liu, S. Shoushtari, Y. Wang, U. S. Kamilov.
  Explored physical model-based video diffusion models for imaging inverse problems, such as denoising, deblurring, and spatial-temporal interpolation, with applications to face video restoration.
  Website Paper

- **Efficient Model-Based Deep Learning via Network Pruning and Fine-Tuning.** *Journal of Mathematical Imaging and Vision.*
  C. Park, W. Gan, **Z. Zou**, Y. Hu, Z. Sun, and U. S. Kamilov.
  Applied structured pruning and fine-tuning to accelerate model-based image reconstruction with minimal loss in reconstruction quality.
  Paper

- **Deep Equilibrium Learning of Explicit Regularization Functionals for Imaging Inverse Problems.** *IEEE Open Journal of Signal Processing.*
  Z. Zou, J. Liu, B. Wohlberg and U. S. Kamilov.
  Proposed "ELDER", a deep equilibrium regularizer that learns explicit regularization functionals, combining the effectiveness of deep learning with the convergence guarantees of optimization theory.
  Website Paper

- **Robustness of Deep Equilibrium Architectures to Changes in the Measurement Model.** *ICASSP 2023.*
  J. Hu, S. Shoushtari, **Z. Zou**, J. Liu, Z. Sun and U. S. Kamilov.
  Investigated the robustness of deep equilibrium models for imaging inverse problems under physical model changes.
  Paper

EXPERIENCE

PhD Student, University of North Carolina, Chapel Hill — 2024 – Present · Chapel Hill, NC

Intern Researcher, WISION Lab, University of Wisconsin-Madison — May - Aug 2026 · Madison, WI

Research Assistant, Washington University in St. Louis — 2023 – 2023 · St. Louis, MO

Teaching Assistant, Washington University in St. Louis — 2023 – 2023 · St. Louis, MO

Embedded System Engineer, Techphant. Inc — 2020 – 2021 · Guangzhou, China

ACADEMIC SERVICE

Reviewer, ISMAR 2025 (IEEE International Symposium on Mixed and Augmented Reality)

SKILLS

Languages: Chinese (Native), English

Technical: Python, C/C++/CUDA, Blender (renderer)